VTPEH 6270 - Check Point 02

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1. Introduction

1.1 Background

My backgroud is Nutrition and my concentration is FSH. Also, I have a huge interest in Psychology. So I'm always wondering if there is a way to study the relationship between nutrition and mental health. NHANES provides nationally representative data that make it possible for me to explore how dietary behaviors could relate to people's mental health in the United States.

1.2 Project Question

Is higher total daily energy intake associated with more frequent depressive symptoms?

1.3 Data set overview

The data used is from the NHANES 2021 to 2023 data, conducted by the CDC of United States. The study population includes children and adults from non institutionalized U.S. population who participated in interviews and examinations conducted as part of NHANES.

2. Data Set Exploration

2.1 Loading Data

```r
library(tidyverse)

# Set the working direction
setwd("/Users/shiyi11/Desktop/R/6270/6270 Check Point 02")

# Import the data set as a data frame
data_nh = read.csv("nhanes_L_merged_diet.csv")

glimpse(data_nh)
```

2.2 Variable Selection

```r
# Calculate total depression score
# Make sure that each question is scored 0-3, total score ranges from 0-27
dpq_items = c("DPQ010","DPQ020","DPQ030","DPQ040","DPQ050",
               "DPQ060","DPQ070","DPQ080","DPQ090")

data_nh = data_nh %>%
  # Select 0-3 scores only
  mutate(across(all_of(dpq_items),
                ~ ifelse(.x %in% 0:3, .x, NA))) %>%
  # Calculate the final score
  mutate(Depression_Score = rowSums(across(all_of(dpq_items)), na.rm = FALSE))


# Select only the variables needed for this analysis
data_nhdm = data_nh %>%
  select(
    # ID variable
    SEQN, # Respondent ID
    # Demographics
    RIAGENDR, # Gender (1=Male, 2=Female)
    RIDAGEYR, # Age in years
    # Depression condition
    Depression_Score,
    # Nutrition variables
    DR1TKCAL, # Total calories (kcal)
    DR1TPROT, # Total protein (g)
    DR1TCARB, # Total carbohydrate (g)
    DR1TTFAT  # Total fat (g)
  )

# Keep only complete cases
data_nhdm = data_nhdm %>%
  filter(
    !is.na(DR1TKCAL),
    !is.na(DR1TPROT),
    !is.na(DR1TCARB),
    !is.na(DR1TTFAT),
    Depression_Score >= 0
    )
```

```r
nrow(data_nhdm)
ncol(data_nhdm)
unique(data_nhdm$Depression_Score)

# Export the new csv file with fewer variables
write_csv(data_nhdm, "nhanes_L_diet_depression.csv")
```

2.3 Variable Overview

```r
# Convert the gender data to factor
data_nhdm <- data_nhdm %>%
  mutate(
    RIAGENDR = factor(
      RIAGENDR,
      levels = c(1, 2),
      labels = c("Male", "Female")
    )
  )

# Create a table describing the variables
vars = c("SEQN", "RIAGENDR", "RIDAGEYR", "Depression_Score",
          "DR1TKCAL", "DR1TPROT", "DR1TCARB", "DR1TTFAT")
variable_info = data.frame(
  Variable = c("SEQN", "RIAGENDR", "RIDAGEYR", "Depression_Score",
               "DR1TKCAL", "DR1TPROT", "DR1TCARB", "DR1TTFAT"),
  Description = c(
    "Respondent ID number",
    "Gender",
    "Age in years",
    "Depression score",
    "Total daily energy intake",
    "Total daily protein intake",
    "Total daily carbohydrate intake",
    "Total daily fat intake"
  ),
  Type = c("Discrete", "Categorical", "Continuous", "Continuous",
           "Continuous", "Continuous", "Continuous", "Continuous"),
  Class = sapply(data_nhdm[, vars], function(x) class(x)[1]),
  Unit = c("ID", "1=Male, 2=Female", "years", "0-27 points",
           "kcal", "g", "g", "g"),
  row.names = NULL
)
```

```r
# Display as a formatted table
library(kableExtra)
kable(variable_info,
      caption = "Overview of Selected Variables",
      booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped",
                                  "hold_position",
                                  "left"))
```

Table 1: Overview of Selected Variables

Variable	Description	Type	Class	Unit
SEQN	Respondent ID number	Discrete	integer	ID
RIAGENDR	Gender	Categorical	factor	1=Male, 2=Female
RIDAGEYR	Age in years	Continuous	integer	years
Depression_Score	Depression score	Continuous	numeric	0-27 points
DR1TKCAL	Total daily energy intake	Continuous	integer	kcal
DR1TPROT	Total daily protein intake	Continuous	numeric	g
DR1TCARB	Total daily carbohydrate intake	Continuous	numeric	g
DR1TTFAT	Total daily fat intake	Continuous	numeric	g

2.4 Data Visualization

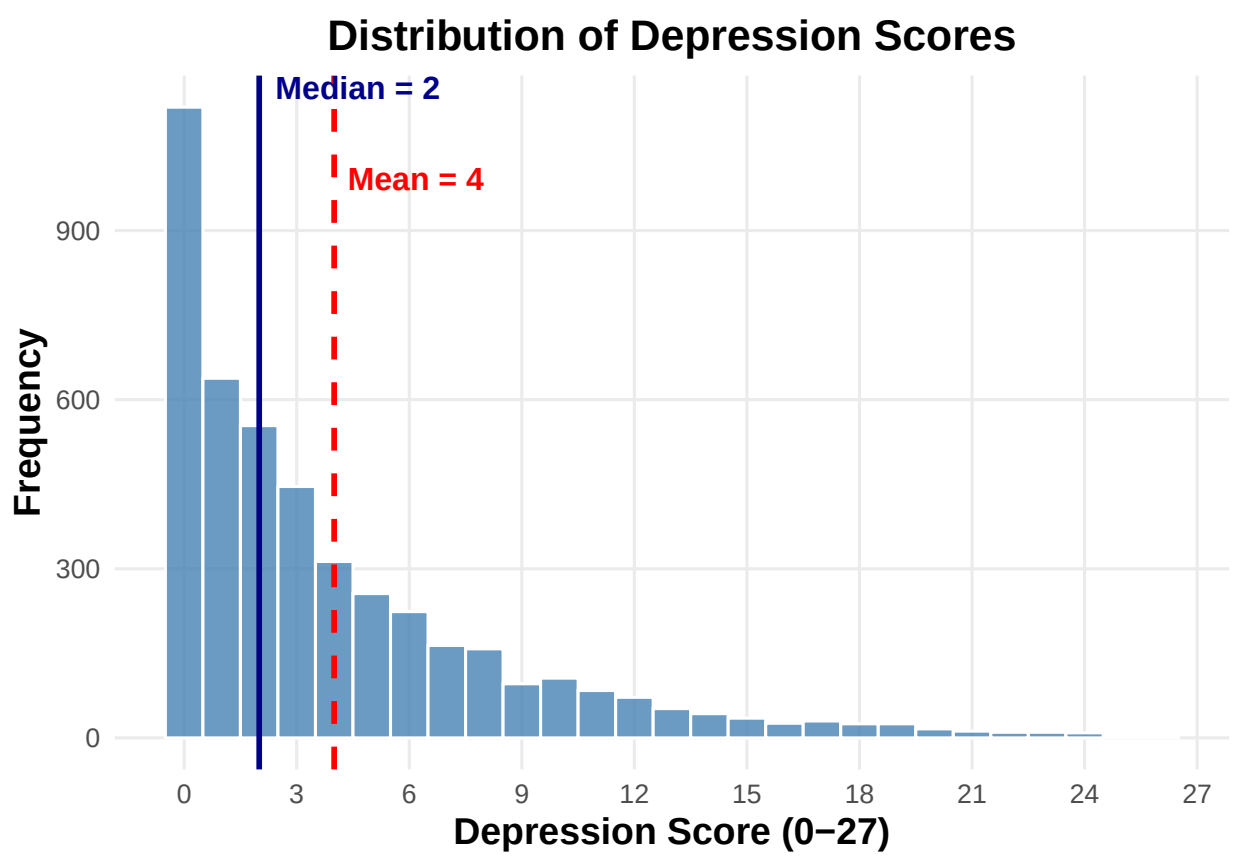
2.4.1 Histogram

I choose to use the depression score to produce a histogram to show the distribution.

```r
# Calculate the mean and median
mean_score = mean(data_nhdm$Depression_Score, na.rm = TRUE)
median_score = median(data_nhdm$Depression_Score, na.rm = TRUE)

# Create histogram
ggplot(data_nhdm, aes(x = Depression_Score)) +
  # Add histogram
  geom_histogram(
    binwidth = 1,
    boundary = -0.5,
    fill = "steelblue",
    color = "white",
    alpha = 0.8
  ) +
  # Add mean line
  geom_vline(
    aes(xintercept = mean_score),
    color = "red",
    linetype = "dashed",
    linewidth = 1
  ) +
  # Add median line
  geom_vline(
    aes(xintercept = median_score),
    color = "darkblue",
    linetype = "solid",
    linewidth = 1
  ) +
  # Add annotations
  annotate("text", x = mean_score, y = Inf,
           label = paste("Mean =", round(mean_score, 2)),
           vjust = 5,
           hjust = -0.1,
           color = "red", fontface = "bold") +
```

```r
  annotate("text", x = median_score, y = Inf,
           label = paste("Median =", round(median_score, 2)),
           vjust = 1,
           hjust = -0.1,
           color = "darkblue", fontface = "bold") +
  labs(
    title = "Distribution of Depression Scores",
    x = "Depression Score (0-27)",
    y = "Frequency"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 16, hjust = 0.5),
    axis.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank()
  ) +
  scale_x_continuous(breaks = seq(0, 27, 3))
```

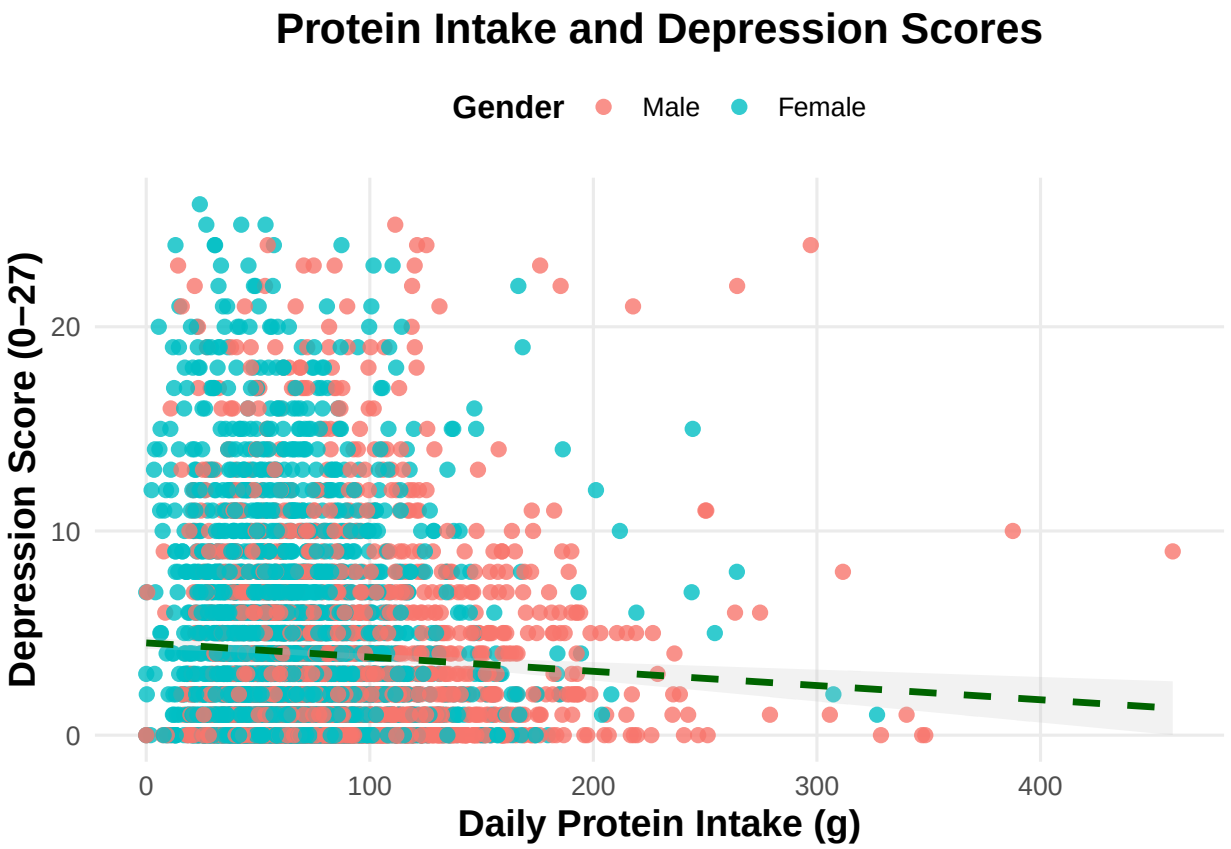


2.4.2 Scatterplot

I choose to use scatterplot to show the relationship between the protein intake and depression score.

```r
# Create scatterplot
ggplot(data_nhdm,
```

```r
      aes(x = DR1TPROT,
          y = Depression_Score,
          color = RIAGENDR)) +
# Add points
geom_point(
  alpha = 0.8,
  size = 2.5,
  shape = 16
) +
# Add trend line
geom_smooth(
  method = "lm",
  se = TRUE,                      # Show confidence interval
  color = "darkgreen",
  fill = "grey",
  linetype = "dashed",
  linewidth = 1.2,
  alpha = 0.2
) +
labs(
  title = "Protein Intake and Depression Scores",
  x = "Daily Protein Intake (g)",
  y = "Depression Score (0-27)",
  color = "Gender"
) +
# Change theme
theme_minimal(base_size = 12) +
theme(
  plot.title = element_text(face = "bold", size = 16, hjust = 0.5),
  axis.title = element_text(face = "bold", size = 14),
  legend.position = "top",
  legend.title = element_text(face = "bold"),
  panel.grid.minor = element_blank()
)
```



3. AI Use Disclosure Statement

This document was generated using ChatGPT to draft some of the rough code, explain R functions, and assist with improving and debugging R code. All analysis, interpretations, and final written answer were reviewed and completed by the author.